

Medical Consultation System

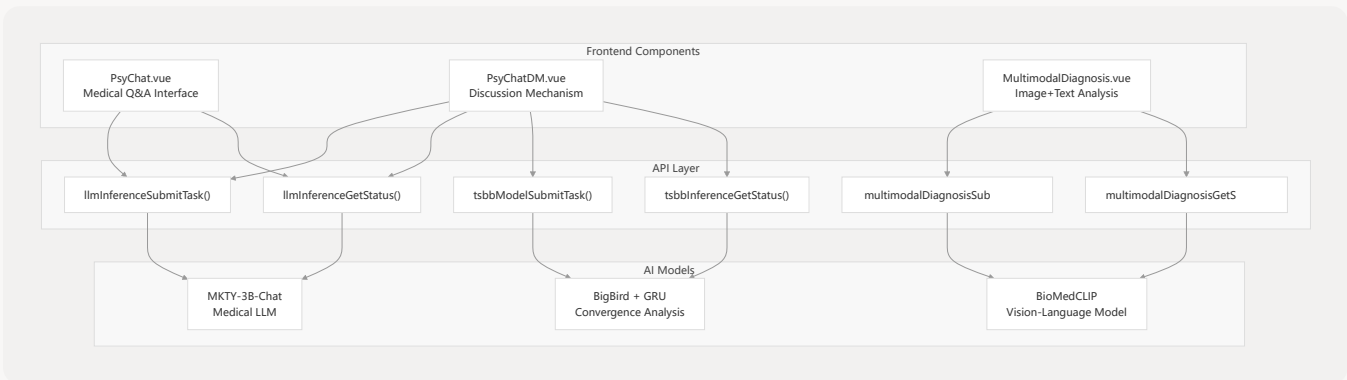
> Relevant source files

This document covers the AI-powered medical consultation system, which provides intelligent Q&A capabilities, multimodal diagnosis, and discussion mechanisms for medical decision-making. The system integrates large language models, computer vision, and time series analysis to assist healthcare professionals and patients.

For information about medical records management and task workflows, see [Medical Records & Task Management](#). For backend API implementation details, see [Core API Endpoints](#). For AI model services, see [AI & Machine Learning Services](#).

System Overview

The Medical Consultation System comprises three primary consultation modalities: direct medical Q&A, multi-agent discussion mechanisms, and multimodal diagnostic analysis. Each modality leverages different AI models and provides distinct interaction patterns suited for various medical consultation scenarios.



Sources: frontend/src/components/modules/PsyChat.vue 1-793

frontend/src/components/modules/PsyChatDM.vue 1-827

frontend/src/components/modules/MultimodalDiagnosis.vue 1-435

Core Consultation Interfaces

Medical Q&A Interface (PsyChat)

The `PsyChat.vue` component provides direct medical question-answering capabilities using the MKTY-3B-Chat model. The interface supports conversational interactions with session persistence

and RAG (Retrieval Augmented Generation) integration.

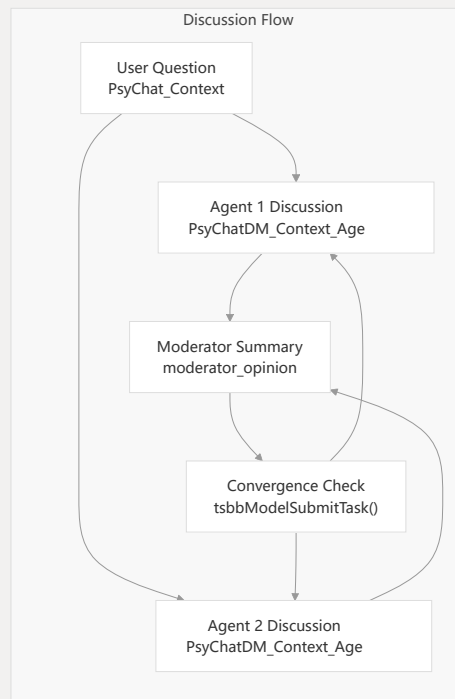
Feature	Implementation	API Functions
Chat Interface	Vue.js reactive chat UI	<code>llmInferenceSubmitTask()</code> , <code>llmInferenceGetStatus()</code>
Session Management	Local storage + backend persistence	<code>saveLlmSession()</code> , <code>getLlmSession()</code> , <code>deleteLlmSession()</code>
RAG Integration	Knowledge base selection + search	<code>searchKnowledgePieces()</code> , <code>getUserFavorites()</code>
Export Functionality	PDF generation	<code>exportChatToPDF()</code>
Real-time Updates	Polling every 2.5 seconds	Status polling interval

The chat interface maintains conversation context in `PsyChat_ChatArr` with roles `'user'` , `'assistant'` , and `'knowledge_base'` for RAG content.

- Sources:
- `frontend/src/components/modules/PsyChat.vue` 37-40
 - `frontend/src/components/modules/PsyChat.vue` 51-128
 - `frontend/src/components/modules/PsyChat.vue` 145-160

Multi-Agent Discussion Mechanism (PsyChatDM)

The `PsyChatDM.vue` component implements an advanced discussion system where multiple AI agents debate medical questions before reaching consensus. This mechanism provides deeper analysis through structured multi-round discussions.



Hyperparameters

PsyChatDM_HyperParameters_AgentNumber

PsyChatDM_HyperParameters_Epoch

PsyChatDM_HyperParameters_ConvergenceThreshold

The discussion process follows this algorithm:

1. Initialize agents and discussion parameters
2. For each epoch (round):
 - Each agent provides analysis based on previous moderator feedback
 - Moderator synthesizes agent perspectives
3. Calculate convergence using TSBB model analysis
4. Save complete discussion session

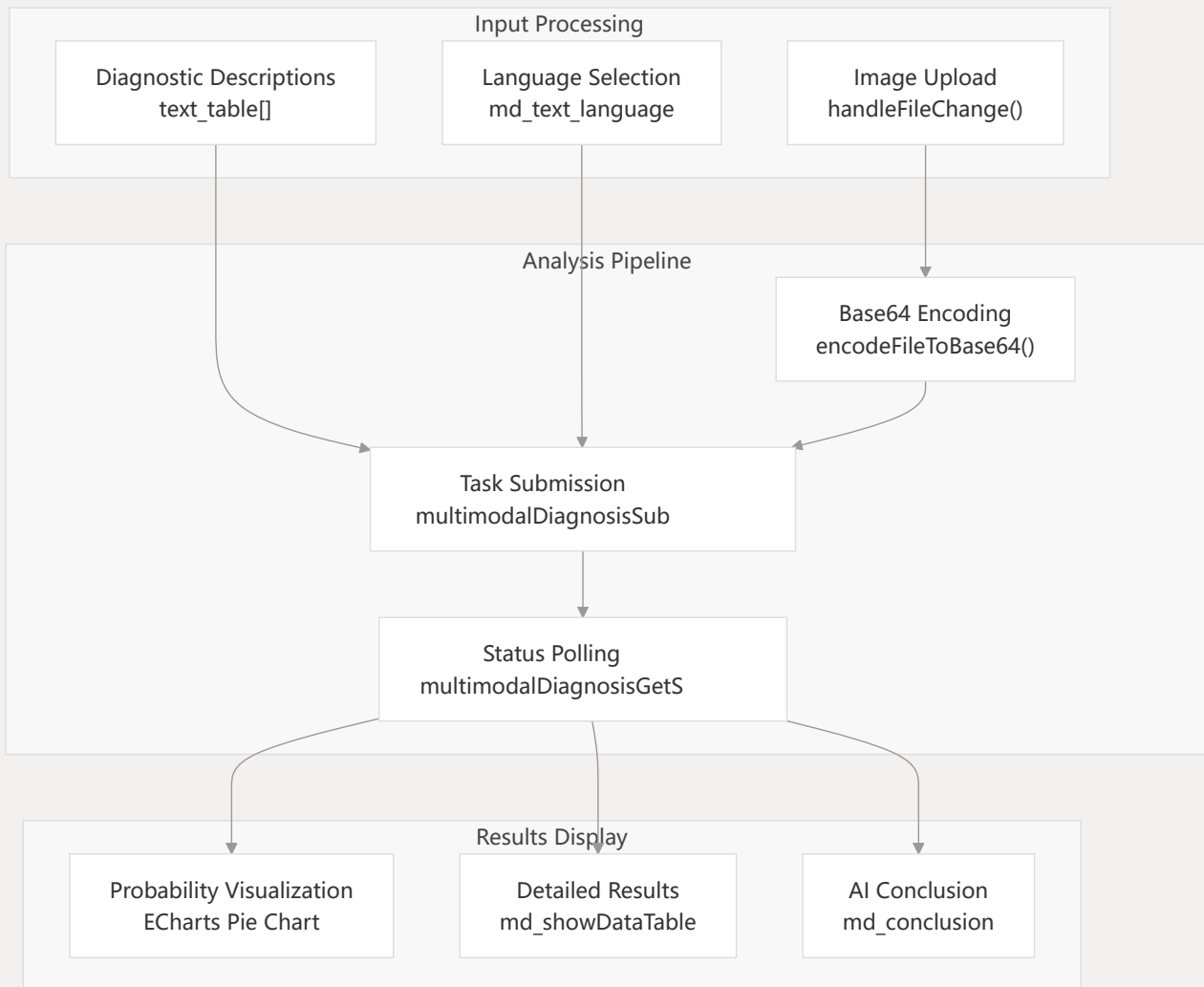
Sources: frontend/src/components/modules/PsyChatDM.vue 87-152

frontend/src/components/modules/PsyChatDM.vue 177-201

frontend/src/components/modules/PsyChatDM.vue 58-62

Multimodal Diagnostic Analysis

The **MultimodalDiagnosis.vue** component combines medical imaging with textual descriptions to provide diagnostic probability distributions using the BioMedCLIP model.



The multimodal analysis supports:

- Medical image formats: CT, MRI, X-ray, etc.
- Bilingual text input (Chinese/English with automatic translation)
- Probability distribution visualization using ECharts
- Detailed probability tables with English labels

Sources: frontend/src/components/modules/MultimodalDiagnosis.vue 144-191

frontend/src/components/modules/MultimodalDiagnosis.vue 124-143

frontend/src/components/modules/MultimodalDiagnosis.vue 87-123

Session Management and Persistence

All consultation interfaces implement comprehensive session management for maintaining conversation history and enabling collaborative medical workflows.

Component	Session Type	Storage Format	Key Functions
PsyChat	Standard Chat	JSON array of role/content objects	<code>saveLlmSession(sessionId, chatArr, 0)</code>
PsyChatDM	Discussion	Array with question + step results	<code>saveLlmSession(-1, stepResults, 1)</code>
MultimodalDiagnosis	Analysis Results	Structured diagnostic data	Results displayed in modal dialogs

Session management includes:

- Automatic saving after successful AI responses
- Session listing with timestamps via `getLlmSessionList()`
- Session deletion with confirmation prompts
- PDF export functionality for chat sessions

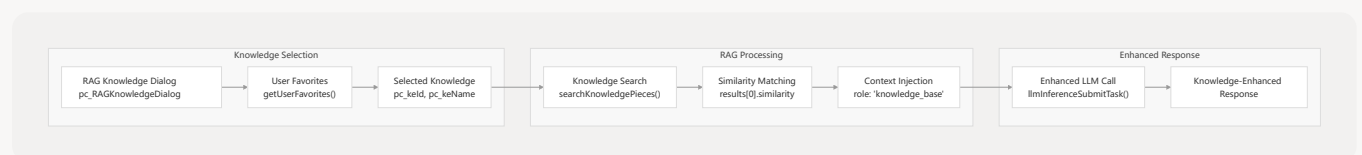
Sources: `frontend/src/components/modules/PsyChat.vue` 110–118

`frontend/src/components/modules/PsyChatDM.vue` 141–149

`frontend/src/components/modules/PsyChat.vue` 200–221

RAG Integration and Knowledge Enhancement

The medical Q&A interface supports Retrieval Augmented Generation through integration with user-favorited knowledge entities.



RAG integration process:

1. User selects knowledge entity from favorites
2. System searches for relevant knowledge pieces using question similarity
3. Best matching knowledge piece is injected into conversation context
4. LLM generates response with enhanced knowledge context

Sources: `frontend/src/components/modules/PsyChat.vue` 63–79

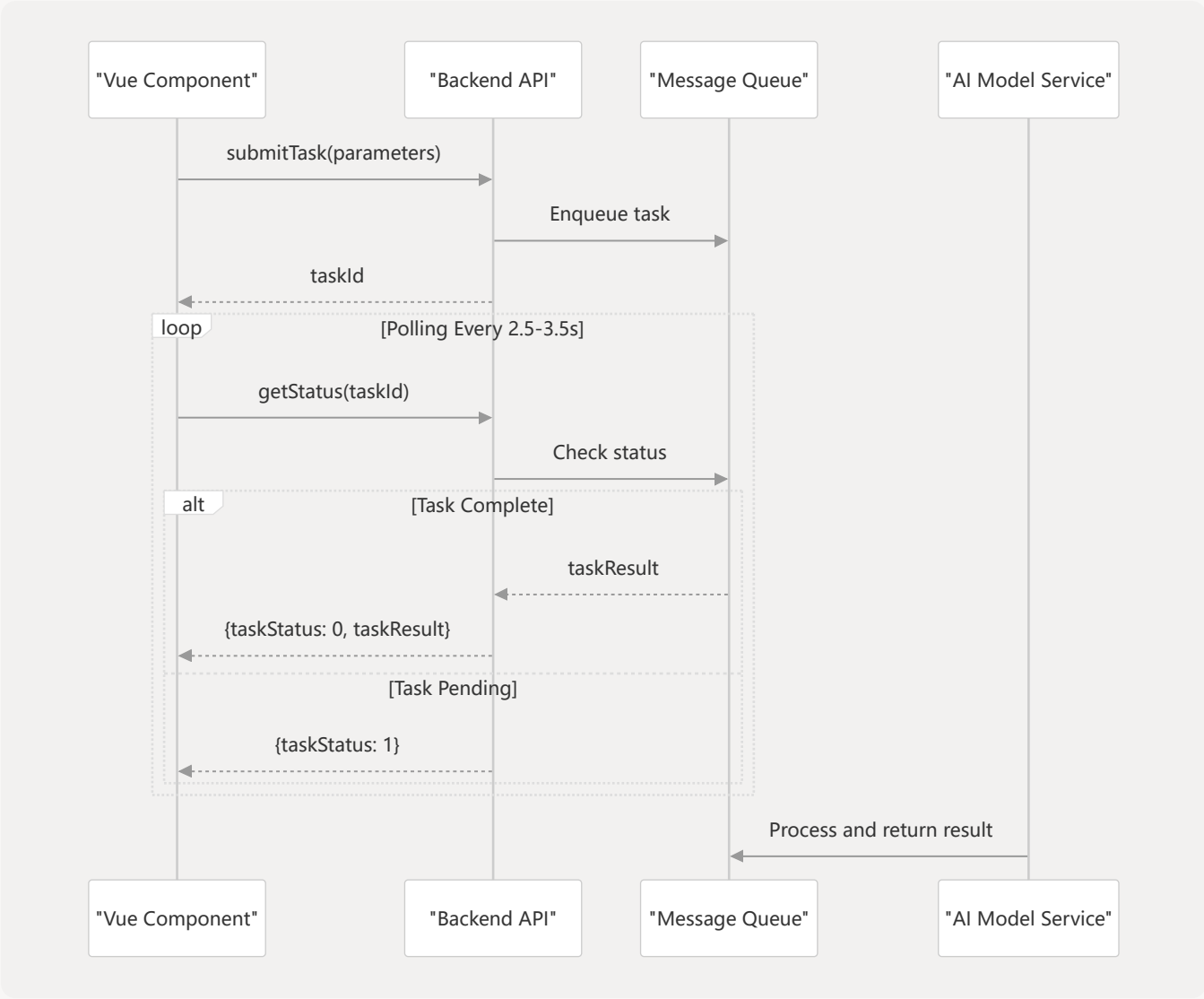
`frontend/src/components/modules/PsyChat.vue` 222–230

`frontend/src/components/modules/PsyChat.vue` 338–362

Asynchronous Task Processing

All AI consultation features use asynchronous task processing with polling-based status checking to handle the computational requirements of AI model inference.

Task Submission and Polling Pattern



Polling Intervals by Component

Component	Polling Interval	Task Type	Status Function
PsyChat	2500ms	LLM Inference	<code>llmInferenceGetStatus()</code>
PsyChatDM	2555ms (LLM), 2567ms (TSBB)	Multi-modal	<code>llmInferenceGetStatus()</code> , <code>tsbbInferenceGetStatus()</code>
MultimodalDiagnosis	3210ms	Vision-Language	<code>multimodalDiagnosisGetStatus()</code>

Sources: frontend/src/components/modules/PsyChat.vue 87-121

frontend/src/components/modules/PsyChatDM.vue 153-176

frontend/src/components/modules/MultimodalDiagnosis.vue 160-185

Error Handling and User Experience

The consultation system implements comprehensive error handling and user feedback mechanisms to ensure reliable medical consultations.

Loading States and User Feedback

Each component manages loading states during AI processing: